



Technical Service Bulletin: TSB150011

Released Date: 21-Mar-2017

Thermal Recirculation Valve Introduction

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- ISX15 CM2250
- ISX15 CM2250 SN
- ISX15 CM2350 X101
- QSX15 CM2250 ECF
- QSX15 CM2250 X115
- QSX15 CM2350 X105
- QSX15 CM2350 X106
- X15 CM2350 X114B
- X15 CM2350 X116B

Description of Change

The purpose of this document is to notify the field of thermal recirculation valve options available. Thermal recirculation valves were released to improve resistance to fuel filter gelling during engine operation in cold weather conditions.

Thermal Recirculation Valve Overview

- The thermal recirculation valve functions by recirculating warm return fuel from various fuel system components into cold fuel tank(s), and fuel supply plumbing.
- Fuel recirculation occurs when supply fuel temperature is below 25°C [77°F].
- There are two thermal recirculation valve options:
 - There is an up-fit thermal recirculation valve kit, Part Number 4376341, available post production for customers. See Installation Instruction, Thermal Recirculation Valve Installation Instructions, Bulletin 4908442. For up-fit thermal recirculation valve option, see Figure 1 below.
 - Engines built January 2017 and later are equipped with a production fuel drain manifold with integrated thermal recirculation valve, Part Number 3691154. Routing to and from thermal recirculation valve is optional and dependent on original equipment

manufacturer (OEM) specifications. For installation instructions, see associated procedure 006-021 in Section 6. For thermal recirculation valve drain manifold option, See Figure 2.

- Thermal recirculation valve is an optional fuel routing component that can be bypassed if desired. Engine performance is **not** affected if thermal recirculation valve is bypassed.

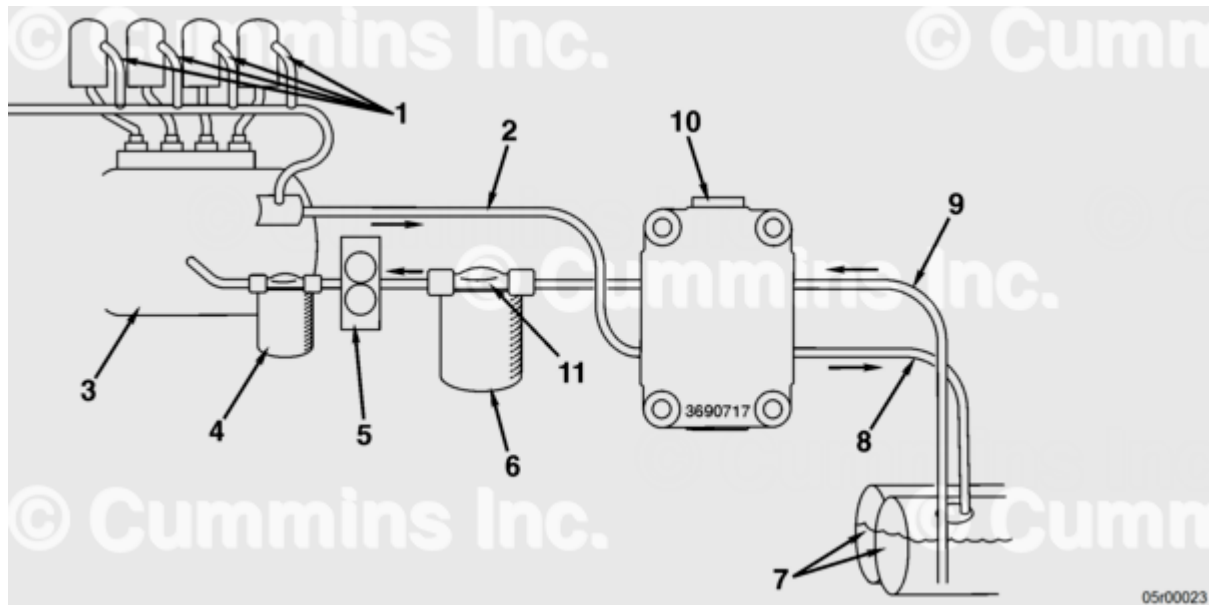


Figure 1, Up-fit Thermal Recirculation Valve Diagram.

1. Injectors
2. Fuel return
3. Fuel pump
4. Secondary fuel filter
5. Fuel transfer pump
6. Primary fuel filter
7. Fuel tanks
8. Return to fuel tank
9. Cold fuel inlet
10. Thermal recirculation valve



Figure 2, Production Thermal Recirculation Valve Diagram.

1. Pressure limiting valve drain
2. Injector drain
3. Thermal recirculation valve drain manifold
4. Drain to tank port
5. High pressure pump drain

Service Parts Availability

Service parts are available. See Table 1 for part numbers.

Table1, Service Parts

Engine Description	Part Description	Part Number
ISX15 CM2250	Thermal Recirculation Valve Up-Fit Kit	4376341
ISX15 CM2250 SN	Thermal Recirculation Valve Up-Fit Kit	4376341
ISX15 CM2350 X101	Thermal Recirculation Valve Up-Fit Kit	4376341
QSX15 CM2250 ECF	Thermal Recirculation Valve Up-Fit Kit	4376341
QSX15 CM2250 X115	Thermal Recirculation Valve Up-Fit Kit	4376341
QSX15 CM2350 X105	Thermal Recirculation Valve Up-Fit Kit	4376341

Table1, Service Parts		
Engine Description	Part Description	Part Number
QSX15 CM2350 X106	Thermal Recirculation Valve Up-Fit Kit	4376341
X15 CM2350 X114B	Fuel Drain Manifold (with integrated thermal recirculation valve)	3691155
X15 CM2350 X116B		

Document History

Date	Details
2015-2-6	Module Created
2015-3-4	Minor updates to clarify supply vs. return ports. Added note about bubbling at primary fuel filter. Updated Figure 1 and 4.
2017-3-15	Revised document to be Introduction of New Thermal Recirculation Valve Options.